

C0 Act Aggregator — Method Comparison

Aggregating C1 measure names into act-level canonical clusters

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Framing

C1 v0.7.0 extracts a `measure_name` string per chunk $\times$ measure, but the same fiscal act surfaces under many different forms (“ERTA”, “Economic Recovery Tax Act of 1981”, “Economic Recovery Tax Act (ERTA)”, “the 1981 tax cut”). Downstream consumers — C2b motivation, deployment summaries, and any cross-country comparison — need a stable canonical mapping at the **act** level. The job of a “C0” Act Aggregator is to produce that mapping.

The Malaysia EN/BM consistency notebook (malay_consistency.qmd) showed that JW single-linkage at threshold 0.15 fragments acts (especially BM-side, where “Subsidi Bahan Api” + “Subsidi Gula” split off from a single budget envelope), and that this fragmentation propagates downstream: C2b reads asymmetric evidence on the two language sides and apparent disagreement is partly clustering noise. So before designing a C0 codebook we need empirical evidence on which aggregation strategy holds up.

This notebook compares five methods on a label-anchored evaluation set (US C1 outputs joined to Romer & Romer’s 40 gold-standard acts) and an out-of-distribution stress test (Malaysia EN/BM paired pairs).

Scope. Corpus-wide per country (one canonical list across all years of a corpus), not per-document and not cross-country. The C1 deployment runs per-country; C0 will too.

Asymmetric failure cost. Over-merging (collapse two real acts into one cluster) corrupts C2b inputs — bad. Under-merging (one act stays split across two clusters) just produces parallel-but-correct classifications — recoverable downstream. The headline pick should weight precision over recall.

From asymmetric cost to coverage + spurious rate. The operational form of the asymmetric-cost argument depends on what gold the eval scores against. Earlier drafts of this notebook used pairwise Precision/Recall against `c0_eval_gold_pairs` (the chunk-tier-tagged pool described below), with the reading that pairwise precision was costly to lose and pairwise recall recoverable. The current draft scores against Romer & Romer’s 49-act list in `us_shocks.csv` — independent of any pipeline output. The asymmetry translates cleanly: **low spurious rate** (few clusters that match no RR act) takes the place of high precision; **high coverage** (many RR acts hit by some cluster) takes the place of high recall; **low fragmentation** (few clusters per matched RR act) is the new precision-flavoured tiebreaker. §“Why we pivoted away from pairwise P/R” below documents the structural ceiling that forced the change.

Gold partition caveat. The new framework scores against R&R’s act list as if it were ground truth. If R&R’s partition splits something a domain expert would have merged (or vice versa), the metric labels inherit that. The chunk-tier leakage caveat that applied to the old pairwise eval no longer applies (the RR-aligned framework bypasses chunk-tier tagging entirely), but the act-list correctness assumption is the same.

What “win” means. High coverage with low spurious rate, low fragmentation index, and high year alignment (most name-matched clusters also satisfy a ± 2 -year window around the act’s signing year). Metrics reported under two complementary name gates side by side — keyword containment (precision-friendly, immune to name-rewriting) and $JW\text{-min} \leq 0.30$ (continuous, sensitive to abbreviations). The gap between the two surfaces methods that emit RR-style canonical names without doing more aggregation work. Bootstrap 95% CIs from resampling the 49 RR acts; `n_boot` = 1000 in the pipeline.

Descriptive baseline

Table 1 sizes the canonicalization problem at the chunk-measure level. Its five rows describe the C1 v0.7.0 output pool feeding C0:

- **Total FISCAL_MEASURE rows** counts chunk $\times$ measure entries C1 flagged as FISCAL_MEASURE. Because v0.7.0 returns a `measures[]` array per chunk, one chunk can contribute multiple rows.
- **Distinct measure_name strings (any rank)** counts unique surface forms across the full pool, including secondary measures from chunks whose primary measure was something else. This is the maximum pool size C0 would face if it had to canonicalize every name C1 ever emits.
- **Distinct measure_name strings (rank == 1)** counts only the primary surface form per chunk. The codebook contract (`c1_measure_id.yml`) requires the first array element to be the most prominent measure discussed, so `rank == 1` is C1's primary pick per chunk. This is the realistic pool size for a primary-measure-only deployment, which matches the current C2b assumption.
- **Years spanned** is the inclusive range of document years contributing chunks.
- **Distinct doc_id** is the number of source documents represented.

The gap between rows 2 and 3 quantifies how much of the surface-form pool comes from secondary mentions. A small gap means rank-1 carries most of the variance and C0 can be evaluated on the rank-1 subset without much loss. A large gap means non-primary measures contribute substantial unique forms, and C0's effective scope (primary-only vs all) becomes a load-bearing methodological choice. The eval set in §3 is built on `rank == 1` only.

Table 1: Measure-name pool from `c1_s2_results`

Metric	Value
Total FISCAL_MEASURE rows	2941
Distinct measure_name strings (any rank)	1976
Distinct measure_name strings (rank == 1)	294
Years spanned	1946-2007
Distinct doc_id	129

Surface-form variance per gold act — the magnitude of the canonicalization problem. For each gold `act_name` with at least one chunk in `c0_eval_gold_pairs`, count the distinct C1 surface forms (rank=1 only) that map to it.

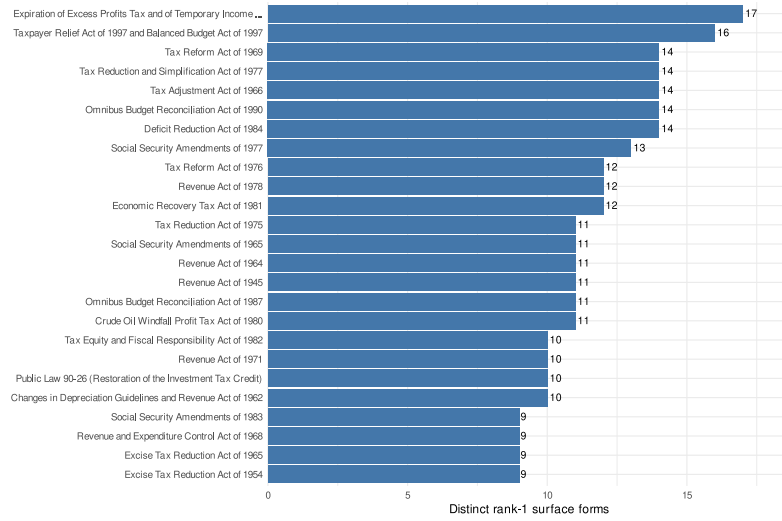


Figure 1: Distinct C1 rank-1 surface forms per gold act (top 25 most fragmented)

Table 2 drills into the same data as Figure 1, using the full any-rank measure pool (rather than the rank-1 subset used by the figure). Gold acts appear as group headers most-fragmented first, annotated with the total number of distinct C1 surface forms that ever appeared in chunks tagged with that act at any rank. Under each group, the top five most-frequently-emitted surface forms with the count of chunk-rows in which C1 emitted them. The shape of the within-group distribution is what C0 has to recover: a dominant first form with a long tail rewards conservative merging because the dominant form is already a serviceable canonical; a flat distribution needs more aggressive merging to collapse parallel forms.

Table 2: Top 5 C1 surface forms per gold act (any rank). Group headers list the total number of distinct surface forms for the act; acts with fewer than five forms show all of them.

Surface form	Appearances
Social Security Amendments of 1977 (946 surface forms)	
Revenue Act of 1964	20
Economic Recovery Tax Act of 1981	18
Social Security Amendments of 1977	18
Tax Reform Act of 1976	18
Social Security Amendments of 1983	17
Revenue Act of 1971 (922 surface forms)	
Revenue Act of 1971	22
Revenue Act of 1950	18
Revenue Act of 1964	16
Social Security Amendments of 1983	15
Tax Reform Act of 1986	14
Revenue Act of 1964 (883 surface forms)	
Revenue Act of 1964	24
Economic Recovery Tax Act of 1981	13
Social Security Amendments of 1983	13
Revenue Act of 1950	11
Excise Tax Reduction Act of 1965	10
Omnibus Budget Reconciliation Act of 1993 (862 surface forms)	
Revenue Act of 1964	20
Revenue Act of 1950	15
Economic Recovery Tax Act of 1981	14
Social Security Amendments of 1983	13
Tax Equity and Fiscal Responsibility Act of 1982	12
Deficit Reduction Act of 1984 (835 surface forms)	
Economic Recovery Tax Act of 1981	21
Social Security Amendments of 1983	21
Deficit Reduction Act of 1984 ⁵	18
Revenue Act of 1950 (811 surface forms)	21
Revenue Act of 1964 (805 surface forms)	21
Revenue Act of 1971 (791 surface forms)	21
Revenue Act of 1986 (785 surface forms)	21
Revenue Act of 1981 (779 surface forms)	21
Revenue Act of 1983 (773 surface forms)	21
Revenue Act of 1980 (767 surface forms)	21
Revenue Act of 1976 (761 surface forms)	21
Revenue Act of 1978 (755 surface forms)	21
Revenue Act of 1975 (749 surface forms)	21
Revenue Act of 1973 (743 surface forms)	21
Revenue Act of 1970 (737 surface forms)	21
Revenue Act of 1968 (731 surface forms)	21
Revenue Act of 1966 (725 surface forms)	21
Revenue Act of 1963 (719 surface forms)	21
Revenue Act of 1961 (713 surface forms)	21
Revenue Act of 1959 (707 surface forms)	21
Revenue Act of 1957 (701 surface forms)	21
Revenue Act of 1955 (695 surface forms)	21
Revenue Act of 1953 (689 surface forms)	21
Revenue Act of 1951 (683 surface forms)	21
Revenue Act of 1949 (677 surface forms)	21
Revenue Act of 1947 (671 surface forms)	21
Revenue Act of 1945 (665 surface forms)	21
Revenue Act of 1943 (659 surface forms)	21
Revenue Act of 1941 (653 surface forms)	21
Revenue Act of 1939 (647 surface forms)	21
Revenue Act of 1937 (641 surface forms)	21
Revenue Act of 1935 (635 surface forms)	21
Revenue Act of 1933 (629 surface forms)	21
Revenue Act of 1931 (623 surface forms)	21
Revenue Act of 1929 (617 surface forms)	21
Revenue Act of 1927 (611 surface forms)	21
Revenue Act of 1925 (605 surface forms)	21
Revenue Act of 1923 (599 surface forms)	21
Revenue Act of 1921 (593 surface forms)	21
Revenue Act of 1919 (587 surface forms)	21
Revenue Act of 1917 (581 surface forms)	21
Revenue Act of 1915 (575 surface forms)	21
Revenue Act of 1913 (569 surface forms)	21
Revenue Act of 1911 (563 surface forms)	21
Revenue Act of 1909 (557 surface forms)	21
Revenue Act of 1907 (551 surface forms)	21
Revenue Act of 1905 (545 surface forms)	21
Revenue Act of 1903 (539 surface forms)	21
Revenue Act of 1901 (533 surface forms)	21
Revenue Act of 1899 (527 surface forms)	21
Revenue Act of 1897 (521 surface forms)	21
Revenue Act of 1895 (515 surface forms)	21
Revenue Act of 1893 (509 surface forms)	21
Revenue Act of 1891 (503 surface forms)	21
Revenue Act of 1889 (497 surface forms)	21
Revenue Act of 1887 (491 surface forms)	21
Revenue Act of 1885 (485 surface forms)	21
Revenue Act of 1883 (479 surface forms)	21
Revenue Act of 1881 (473 surface forms)	21
Revenue Act of 1879 (467 surface forms)	21
Revenue Act of 1877 (461 surface forms)	21
Revenue Act of 1875 (455 surface forms)	21
Revenue Act of 1873 (449 surface forms)	21
Revenue Act of 1871 (443 surface forms)	21
Revenue Act of 1869 (437 surface forms)	21
Revenue Act of 1867 (431 surface forms)	21
Revenue Act of 1865 (425 surface forms)	21
Revenue Act of 1863 (419 surface forms)	21
Revenue Act of 1861 (413 surface forms)	21
Revenue Act of 1859 (407 surface forms)	21
Revenue Act of 1857 (401 surface forms)	21
Revenue Act of 1855 (395 surface forms)	21
Revenue Act of 1853 (389 surface forms)	21
Revenue Act of 1851 (383 surface forms)	21
Revenue Act of 1849 (377 surface forms)	21
Revenue Act of 1847 (371 surface forms)	21
Revenue Act of 1845 (365 surface forms)	21
Revenue Act of 1843 (359 surface forms)	21
Revenue Act of 1841 (353 surface forms)	21
Revenue Act of 1839 (347 surface forms)	21
Revenue Act of 1837 (341 surface forms)	21
Revenue Act of 1835 (335 surface forms)	21
Revenue Act of 1833 (329 surface forms)	21
Revenue Act of 1831 (323 surface forms)	21
Revenue Act of 1829 (317 surface forms)	21
Revenue Act of 1827 (311 surface forms)	21
Revenue Act of 1825 (305 surface forms)	21
Revenue Act of 1823 (299 surface forms)	21
Revenue Act of 1821 (293 surface forms)	21
Revenue Act of 1819 (287 surface forms)	21
Revenue Act of 1817 (281 surface forms)	21
Revenue Act of 1815 (275 surface forms)	21
Revenue Act of 1813 (269 surface forms)	21
Revenue Act of 1811 (263 surface forms)	21
Revenue Act of 1809 (257 surface forms)	21
Revenue Act of 1807 (251 surface forms)	21
Revenue Act of 1805 (245 surface forms)	21
Revenue Act of 1803 (239 surface forms)	21
Revenue Act of 1801 (233 surface forms)	21
Revenue Act of 1799 (227 surface forms)	21
Revenue Act of 1797 (221 surface forms)	21
Revenue Act of 1795 (215 surface forms)	21
Revenue Act of 1793 (209 surface forms)	21
Revenue Act of 1791 (203 surface forms)	21
Revenue Act of 1789 (197 surface forms)	21
Revenue Act of 1787 (191 surface forms)	21
Revenue Act of 1785 (185 surface forms)	21
Revenue Act of 1783 (179 surface forms)	21
Revenue Act of 1781 (173 surface forms)	21
Revenue Act of 1779 (167 surface forms)	21
Revenue Act of 1777 (161 surface forms)	21
Revenue Act of 1775 (155 surface forms)	21
Revenue Act of 1773 (149 surface forms)	21
Revenue Act of 1771 (143 surface forms)	21
Revenue Act of 1769 (137 surface forms)	21
Revenue Act of 1767 (131 surface forms)	21
Revenue Act of 1765 (125 surface forms)	21
Revenue Act of 1763 (119 surface forms)	21
Revenue Act of 1761 (113 surface forms)	21
Revenue Act of 1759 (107 surface forms)	21
Revenue Act of 1757 (101 surface forms)	21
Revenue Act of 1755 (95 surface forms)	21
Revenue Act of 1753 (89 surface forms)	21
Revenue Act of 1751 (83 surface forms)	21
Revenue Act of 1749 (77 surface forms)	21
Revenue Act of 1747 (71 surface forms)	21
Revenue Act of 1745 (65 surface forms)	21
Revenue Act of 1743 (59 surface forms)	21
Revenue Act of 1741 (53 surface forms)	21
Revenue Act of 1739 (47 surface forms)	21
Revenue Act of 1737 (41 surface forms)	21
Revenue Act of 1735 (35 surface forms)	21
Revenue Act of 1733 (29 surface forms)	21
Revenue Act of 1731 (23 surface forms)	21
Revenue Act of 1729 (17 surface forms)	21
Revenue Act of 1727 (11 surface forms)	21
Revenue Act of 1725 (5 surface forms)	21

Table 3: Distribution of surface-form fragmentation across the 39 gold acts

Metric	Value
Acts evaluated	39.00
Median surface forms / act	10.00
Mean surface forms / act	9.54
Max surface forms / act	17.00
Acts with ≥ 5 forms	34.00
Acts with exactly 1 form	1.00

The headline: even with R&R’s well-known and well-named acts, the median act surfaces under multiple forms; the tail is heavy. A C0 that returns one cluster per surface form (i.e., no merging) is useless; a C0 that merges everything is also useless. The methods below try to land somewhere useful.

Within-document name variance: 1980 cross-section

Figure 2 sizes the data reduction happening upstream of the heatmaps below. Stage 1 counts every chunk C1 reviewed in each 1980 document (the input to FISCAL_MEASURE classification), split into the “relevant” chunks where C1 emitted at least one measure and the irrelevant chunks where it did not. Stage 2 keeps only the relevant subset — the same blue block from stage 1 redrawn on its own to make the funnel mouth visible. Stage 3 unfolds the relevant chunks into individual measure occurrences (one row per chunk $\times$ `measure_rank` pair), arranged by rank with the rank-1 layer marked in red. The shape of the funnel is the headline: Budget and Treasury chunks emit ≈ 5 -10 measures each, so the rank-1 red band is a small slice of stage 3 — and that red slice is the pool the JW and embedding heatmaps below operate on. Rank ≥ 2 layers (grey) are out of scope for the heatmaps but are not noise; they record secondary measures C1 surfaced in the same chunk.

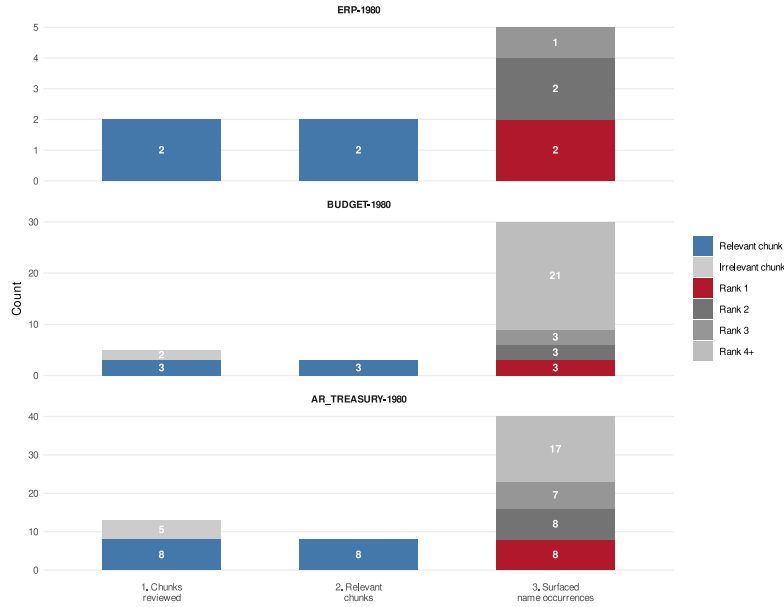
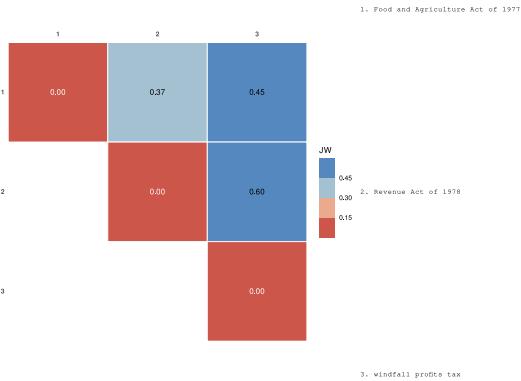


Figure 2: Per-document funnel from C1 input to the rank-1 layer the heatmaps below operate on. Stage 1: chunks C1 reviewed, split into relevant (FISCAL_MEASURE emitted) and irrelevant. Stage 2: the relevant subset alone. Stage 3: all FISCAL_MEASURE occurrences (one per chunk $\times$ `measure_rank` pair) stacked by rank, with rank-1 marked in red. Stages 1-2 are counted in chunks; stage 3 in occurrences (so a single relevant chunk producing five measures contributes 1 to stage 2 and 5 to stage 3). The y-axis is free per facet to keep ERP-1980’s small counts legible alongside Treasury’s larger ones.

Figure 3 zooms into the rank-1 layer of Figure 2: it shows the pairwise Jaro-Winkler distance matrix among those rank-1 C1 measure names within each document (mirroring the Methods 2 + 3 pool used in Figure 4, so the two heatmaps carry the same acts). The heatmap pool is the within-doc surface forms intersected with the corpus-wide rank-1 pool (`rownames(c0_us_embeddings)`) — slightly larger than stage 3’s red layer because it includes a few names that are at rank ≥ 2 in *this* document but were rank-1 in some other document, while excluding rank-1 occurrences whose surface form is shared with another (so the same name shown once). Cells in the bottom band ($JW < 0.15$) would merge under single-linkage at the production threshold. Cells in the middle band ($0.15-0.30$) are near-misses — same measure with verbose-vs-bare phrasing, swapped word order, or abbreviation expansion. Cells in the top band (≥ 0.30) are genuinely distinct measures. Read each heatmap together with its numbered measure list (printed

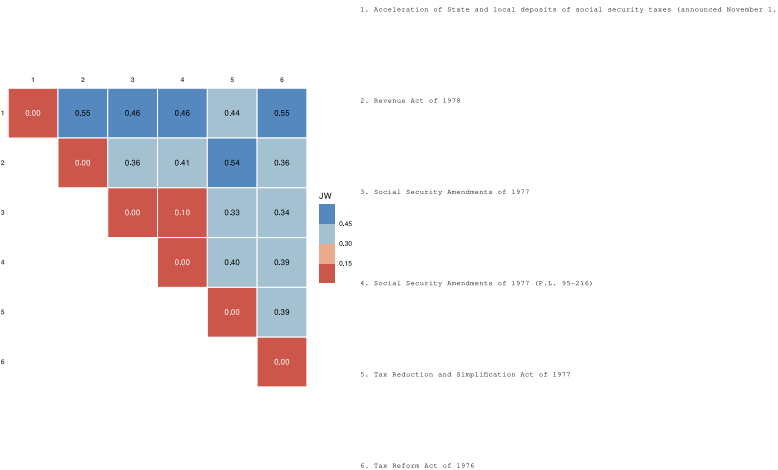
to its right). Off-diagonal cells just above the 0.15 line that link semantically-identical names are the failure cases that drive over-fragmentation; cells far above 0.30 linking surface forms that *are* the same act are the gap JW cannot close on its own.

ERP-1980 — 3 rank-1 C1 measure names (2 non-rank-1 omitted)



(a) Economic Report of the President

BUDGET-1980 — 6 rank-1 C1 measure names (23 non-rank-1 omitted)



(b) Budget

AR_TREASURY-1980 — 10 rank-1 C1 measure names (16 non-rank-1 omitted)

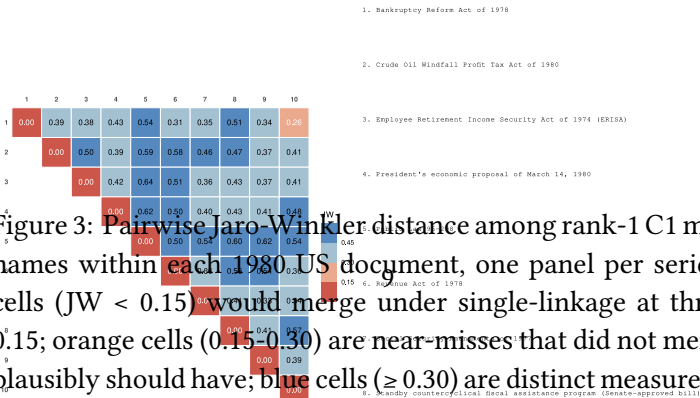
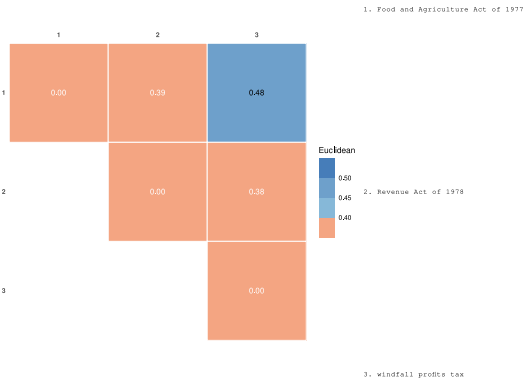


Figure 3: Pairwise Jaro-Winkler distance among rank-1 C1 measure names within each 1980 US document, one panel per series. Red cells ($JW < 0.15$) would merge under single-linkage at threshold 0.15; orange cells ($0.15 \leq JW < 0.30$) are near-misses that did not merge but plausibly should have; blue cells ($JW \geq 0.30$) are distinct measures. Only the upper triangle is shown; the diagonal is $JW = 0$ by construction.

Figure 3 visualised what JW sees within each of the three 1980 US documents. Figure 4 is the same structure on the same documents and the same measure-name pool, but with pairwise *embedding* distance in place of JW. Distances are Euclidean on the L2-normalised E5 vectors `c0_us_embeddings`; because the embeddings are unit-norm, Euclidean distance is a monotonic transform of cosine distance ($d_{\text{euc}} = \sqrt{2 \cdot d_{\text{cos}}}$), so the figure mirrors what Methods 2 + 3 actually cluster on.

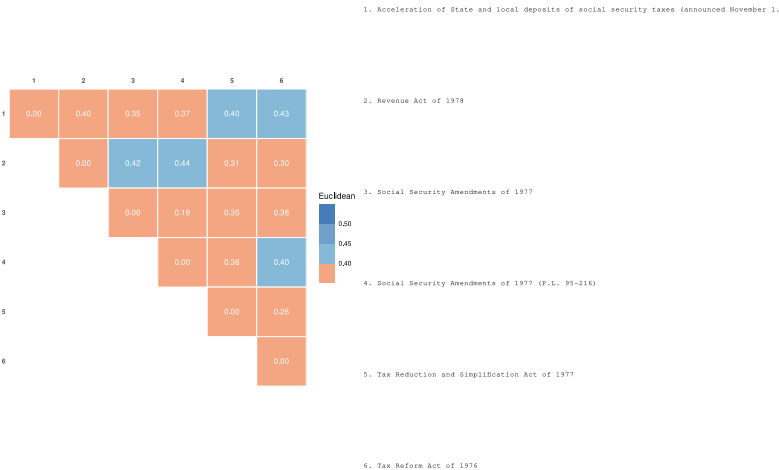
The colour bands are tighter than the JW figure’s because E5 packs everything into a narrow band of the unit sphere: corpus-wide pairwise Euclidean distances sit at median 0.45 with IQR 0.42-0.48. Bands at 0.40 / 0.45 / 0.50 therefore approximate “below corpus IQR” / “within IQR” / “above IQR” / “far above IQR” — a relative-rather-than-absolute reading, in keeping with the verdict text above.

ERP-1980 — 3 embedded C1 measure names (2 without embedding omitted)



(a) Economic Report of the President

BUDGET-1980 — 6 embedded C1 measure names (23 without embedding omitted)



(b) Budget

AR_TREASURY-1980 — 10 embedded C1 measure names (16 without embedding omitted)

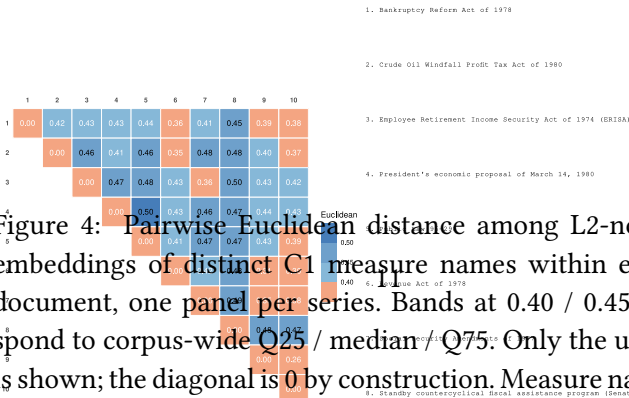


Figure 4: Pairwise Euclidean distance among L2-normalised E5 embeddings of distinct C1 measure names within each 1980 US document, one panel per series. Bands at 0.40 / 0.45 / 0.50 correspond to corpus-wide Q25 / median / Q75. Only the upper triangle is shown; the diagonal is 0 by construction. Measure names without an embedding (i.e., never appeared at rank 1 in the corpus) are

Reading Figure 4 against Figure 3, an immediate conclusion is that imposing the Rank-1 filter as in Figure 4 does not imply losing important fiscal changes. Indeed, doing so, refines the surfaced names and are more in tune with the acts Romer and Romer (n.d.) identify. Importantly, there is little within-document-within act name variance and it could be recuperated both with JW distance and with embeddings.

Tier-tagged eval set (descriptive)

This section documents `c0_eval_gold_pairs`, the chunk-tier-tagged pool used by the embedding-geometry probes in §“Embedding-geometry probe (F16 vs FP32)” and §“Eval-set noise probe”. It is **not** the headline gold for method comparison — that role belongs to R&R’s 49-act list, evaluated under the framework defined in §“Evaluation against RR’s act list” below. The structural reason for the split is in §“Why we pivoted away from pairwise P/R”, which follows immediately.

The eval set projects the chunk-tier system through C1’s LLM output along the chain `aligned_data` → `c1_chunk_data` → `c1_s2_results` → `c0_eval_gold_pairs`:

1. `c1_chunk_data` (built by `R/identify_chunk_tiers.R`) emits `(chunk_id, doc_id, act_name, tier)` rows by string-matching gold passages and act-name subcomponents against chunk text. Tier 1 marks chunks containing a verbatim gold passage; Tier 2 marks chunks matching an act name via subcomponent decomposition or co-occurrence rules. Chunks matching multiple acts within the same tier are row-replicated, one row per `(chunk × act)`. The Tier 1 / Tier 2 exclusion is chunk-level, not `(chunk, act)`-level: any chunk that is Tier 1 for *some* act is barred from Tier 2 consideration for *every* act, so Tier 2 evidence on mixed-tier multi-act chunks is dropped at the source.
2. `c1_s2_results` runs the C1 LLM on these tagged chunks via leave-one-act-out cross-validation; each `(chunk × gold act)` pair is its own fold with its own few-shot set. The LLM’s emitted `measures[]` array is exploded into one row per `(chunk × gold act × measure_rank)`, so a chunk tagged with two gold acts is classified twice and may carry different `measure_name` strings under each gold tag.
3. `build_c0_eval_gold_pairs()` (in `R/c0_aggregator.R`) keeps only rows with `measure_rank == 1`, `pred_label == FISCAL_MEASURE`, and `tier ∈ {1, 2}`; renames `act_name` →

`gold_act_name`; and flags surface forms paired with >1 distinct gold act as `ambiguous` for the evaluator to majority-vote.

Two consequences carry into every downstream metric. (i) The rank-1 filter discards non-primary measures the LLM emitted, so a chunk discussing N gold acts contributes at most N rows with only the rank-1 surface form for each. (ii) Chunks the LLM mislabeled `NOT_FISCAL_MEASURE` drop out entirely, silently truncating multi-act representation whenever the LLM’s per-fold output is inconsistent.

Table 4: Eval-set composition. Rows count (`measure_name`, `gold_act_name`, `chunk_id`) tuples; distinct surface forms count unique `measure_name` strings — the population the clusterer canonicalizes. The two indented rows partition the Tier 1 + Tier 2 pool by whether each surface form maps to one (unambiguous) or more (ambiguous) gold acts. Percentages are shares of the Tier 1 + Tier 2 totals.

Slice	Chunk gold-act rows	× Distinct surface forms	sur- Gold acts cov- ered
Tier 1 (verbatim passage match)	60	54	30
Tier 1 + Tier 2 (full eval pool)	592	267	39
↳ unambiguous names (1 gold act per name)	237 (40%)	206 (77%)	—
↳ ambiguous names (≥2 gold acts per name)	355 (60%)	61 (23%)	—

On ambiguity. Roughly 60% of the Tier 1+2 rows have a `measure_name` that’s been paired with more than one `gold_act_name` across chunks. Most cases trace to Tier 2’s permissive matching: a chunk may be tagged with act X because act X is mentioned somewhere, but the most-prominent measure C1 extracts is a different act. The evaluator handles this by majority-voting per `measure_name` within each bootstrap resample (so the

contingency table operates on unique names). Tier 1 is shown as the cleaner anchor; Tier 1+2 is shown for statistical power.

Gold-set leakage caveat. The Tier 1/Tier 2 tags themselves were assigned by fuzzy-matching the gold `act_name` against the chunk text, so methods that exploit surface similarity (especially JW) will look artificially strong on this evaluation. Treat the per-act recall column as a complement — methods that nail F1 but miss specific acts may be exploiting the leakage.

Why we pivoted away from pairwise P/R

`c0_eval_gold_pairs` has a structural ceiling on pairwise recall that comes from how the pool is constructed, not from any clusterer’s performance. `build_c0_eval_gold_pairs()` assigns *every* rank-1 measure C1 extracted in a chunk to *every* gold act that chunk was tier-tagged with — including chunks where C1’s primary measure was a different act. The act `gold_act_name` becomes a bag of co-occurring rank-1 measures rather than a list of surface variants of one act. Concretely, “Tax Reform Act of 1969” picks up members {ERISA, Asset Depreciation Range (ADR), H.R. 10947, tax bill passed in December} — distinct policy items, not paraphrases. No string-embedding clusterer can or should merge those.

Even on the Tier 1 slice (verbatim-passage tag, supposedly the clean anchor), the same pattern holds. “Deficit Reduction Act of 1984” Tier-1 members include {Economic Recovery Tax Act, Job Training Partnership Act, tax proposals (marginal income tax rate reductions and depreciation reform)}. The Tier 1 filter constrains where C1 emitted its measure, not what measure it emitted.

The §UMAP investigation in the 2026-05-28 commit cycle confirmed this empirically: 16 of 35 eligible Tier 1 acts have top-1 NN same-act rate = 0 on F16 raw embeddings (20/35 under the UMAP precision-friendly winner); across the 162-cell UMAP grid, `cor(% acts reconstructed intact, F1) = +0.82` while `cor(separation, pairwise_recall) = -0.07` — pairwise F1 is essentially a one-variable story about how often a gold act survives HDBSCAN as a single cluster, and that variable is capped by gold-pool construction upstream of the embedding. No UMAP variant strictly dominated the no-UMAP baseline on the original pairwise eval because both hit the same ceiling. See §“Reading the UMAP results” for the full diagnostic.

The pivot. Method comparison now scores against R&R’s authoritative 49-act list in `us_shocks.csv`, which is independent of any pipeline output. The pairwise-P/R evaluation against `c0_eval_gold_pairs` is preserved for the embedding-geometry probes (where median within/between separation is the metric, not a partition recall) and the eval-set noise probe (where the temporal mismatch is the diagnostic) — but no longer drives method selection.

Evaluation against RR’s act list

Reference: what `us_shocks.csv` provides

`c0_us_rr_acts` collapses `us_shocks` (one row per act × quarter × measure_type) to one row per distinct act with `year_signed = year(Date signed)`. 49 acts spanning 1945–2007. The first few rows:

Table 5: First five rows of the RR canonical-act list (`c0_us_rr_acts`), sorted by `year_signed`.

act_id	act_name	year_signed
1	Revenue Act of 1945	1945
2	Social Security Amendments of 1947	1947
3	Revenue Act of 1948	1948
4	Revenue Act of 1950	1950
5	Social Security Amendments of 1950	1950

This list is the metric’s reference: a predicted cluster is “good” iff some member surface form (and the cluster’s representative year) matches an entry in this list under the gates below.

Matching gates

Two complementary name gates, evaluated side by side in every metric table that follows. Both combined with a year window.

Keyword containment (precision-friendly). For each RR act, decompose the act name via `generate_subcomponents()` (the function `R/identify_chunk_tiers.R` already uses for Tier 2 chunk tagging — same decomposition, no new code path) into a set of searchable terms: the full canonical name, “and”-splits, parentheses, embedded “{X} Act of {YYYY}” patterns, Public Law numbers, action+subject phrases. The shared `BROAD_EXCLUSIONS` list strips overly generic fragments. A cluster keyword-matches an RR act

iff *any* of its surface forms (after `squish_for_matching()` normalisation) contains *any* of the act's subcomponent terms. Property: immune to name-rewriting — a method that re-emits the canonical act name as cluster representative gets no extra credit because the test runs against the cluster's *members*, not its canonical.

JW-min (continuous, abbreviation-sensitive). For each (cluster, RR act) pair, compute `min` over cluster members of `stringdist::stringdist(member, rr_act_name, method = "jw")` on lowercased strings. Threshold ≤ 0.30 . Property: catches near-canonical matches that keyword-containment misses (e.g. typographic variants); also catches *false* matches across acts whose names share rare tokens. The JW gate is the relaxed one — it rewards methods that emit RR-style canonical names because those land at JW = 0 against the act.

The gap between coverage under keyword and coverage under JW is itself diagnostic: a method whose keyword and JW coverage agree is recovering the canonical forms via aggregation; a method with much higher JW coverage than keyword coverage is mostly re-emitting canonical names without aggregating much.

Year window

Cluster representative year = median of years extracted from members' surface forms via `extract_year_from_string()` (regex `\b(19|20)\d{2}\b`), falling back to the mode of `doc_year` from `c0_us_measure_pool` when no member name carries an extractable year. Year gate: $|\text{cluster_year} - \text{year_signed}| \leq 2$. Width ± 2 chosen to (i) align with the Method 3 year-blocking variants so methods that respect the year prior get credit, and (ii) absorb retrospective references whose mode `doc_year` lags the signing year by 1–2.

Joint match gate per name gate: name match AND year match. A cluster that name-matches an act but is 30 years off does *not* match.

Headline metrics

Under each name gate, four numbers per variant:

- **Coverage** = $|\{\text{RR acts with } \geq 1 \text{ matching cluster}\}| / 49$. Recall-flavoured.
- **Spurious rate** = $|\{\text{clusters with no RR-act match}\}| / n_{\text{clusters}}$. Precision-flavoured; this is the metric the asymmetric-cost framing weights most.

- **Fragmentation index** = mean over matched RR acts of (# distinct clusters claiming that act). ≥ 1 by construction. Equals 1 when each matched act has exactly one cluster; grows when methods fragment acts.
- **Year alignment** = of name-matched pairs (name gate only, ignoring the year gate), share where the year gate also passes. Tests whether name and year signals agree — low values under the JW gate indicate the JW threshold is producing cross-decade false positives.

Coverage and spurious rate are the headline pair. Fragmentation is the precision-flavoured tiebreak between methods at similar coverage. Year alignment is mostly diagnostic.

Bootstrap CI

Resample the 49 RR acts with replacement, recompute the four metrics, take the 0.025 and 0.975 percentiles. `n_boot = 1000` in the pipeline (the smoke test uses 100 for speed). The bootstrap perturbs *which* acts the method is held accountable for, which is the noise the CI should capture (acts vary in difficulty for any given clusterer). It does *not* perturb the cluster set — clusters are deterministic outputs of each method $\times$ hyperparameter cell.

Worked example

Table 6 demonstrates the matching gates on one act and two contrasting clusters. The first cluster contains the canonical surface form ([Economic Recovery Tax Act of 1981](#)) and matches under both gates. The second is a hypothetical singleton cluster of just [ERTA](#): the keyword check fails (the subcomponent term is the full canonical name; “erta” does not contain “economic recovery tax act of 1981”), and the JW distance from “erta” to the canonical name is well above 0.30. An ERTA-only cluster therefore receives no credit — the method must merge ERTA with the canonical form for either gate to pass, which is exactly the aggregation behaviour C0 is meant to demonstrate.

Table 6: Worked example: matching two hypothetical clusters against *Economic Recovery Tax Act of 1981* under the keyword and JW gates.

Cluster member	Subcomponent terms	Keyword check	JW distance	$JW \leq 0.30$
Economic Recovery Tax Act of 1981	Economic Recovery Tax Act of 1981	PASS	0.00	PASS
ERTA	Economic Recovery Tax Act of 1981	FAIL	0.48	FAIL

Method 1 – JW single-linkage

Before reading the threshold sweep, Figure 5 shows what JW is being asked to recover. Each row is a gold act with ≥ 2 surface forms in the eval pool; the box and jittered points show the pairwise JW distances among the surface forms tagged to that act. Vertical dashes mark the swept thresholds (0.10–0.30). An act whose distribution sits entirely left of a threshold line is fully recoverable at that threshold; an act whose distances spread past 0.30 cannot be merged without dragging in surface forms from neighbouring acts (i.e., over-merge). Ambiguous names are kept under every gold act they’re tagged to, matching the descriptive intent of the figure rather than the evaluator’s majority-vote.

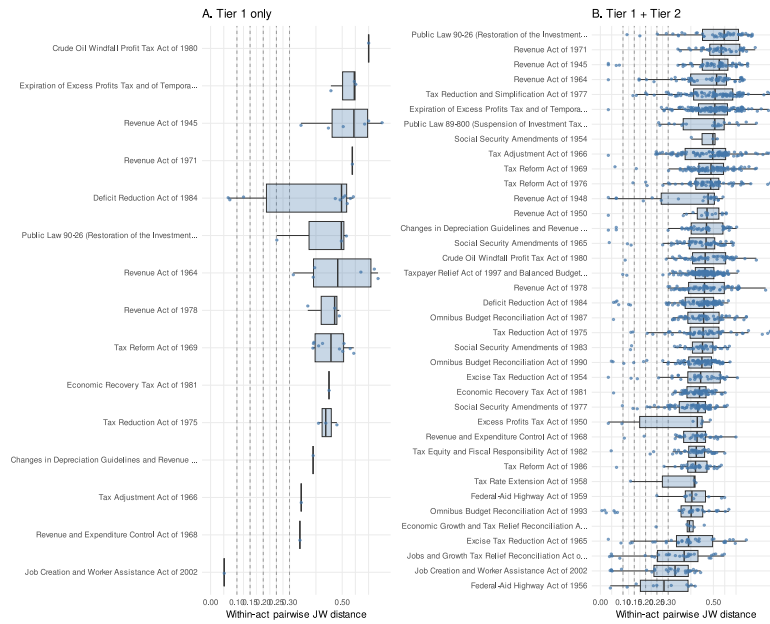


Figure 5: Within-gold-act pairwise JW distances. Vertical dashes mark the swept thresholds. Acts with a single surface form contribute no pairs and are omitted.

What this means for Method 1. The median within-act JW distance is ~ 0.47 on both tiers, while the loosest threshold swept is 0.30. The typical pair of surface forms that *ought* to be merged is therefore geometrically out of reach for JW at any threshold the table considers; only a small minority of pairs (near-duplicates differing in spacing, punctuation, or abbreviation expansion) sit below 0.30. The implication is unavoidable: **JW single-linkage cannot canonicalize this pool on its own.** The threshold sweep in Table 7 below is best read as quantifying the asymmetric coverage/spurious cost given that fundamental ceiling, not as a search for a workable operating point. A method that can bridge [ERTA](#) $\leftrightarrow$ [Economic Recovery Tax Act of 1981](#) — semantic, not lexical — is required, which is what Phase B is for.

Figure 6 checks the same geometry from a different angle: instead of pairwise distances among surface forms, it measures each surface form’s JW distance to its gold-act label. If C0 had the gold canonical list as an oracle and tried to map surface forms to it by JW, the swept thresholds tell us how many surface forms it would catch. The answer mirrors Figure 5: median ~ 0.41 on both tiers, well above 0.30. A handful of exact matches sit at 0 (surface form is the gold label string verbatim), but most surface forms are

too lexically distant from their canonical to be matched without semantic reasoning.

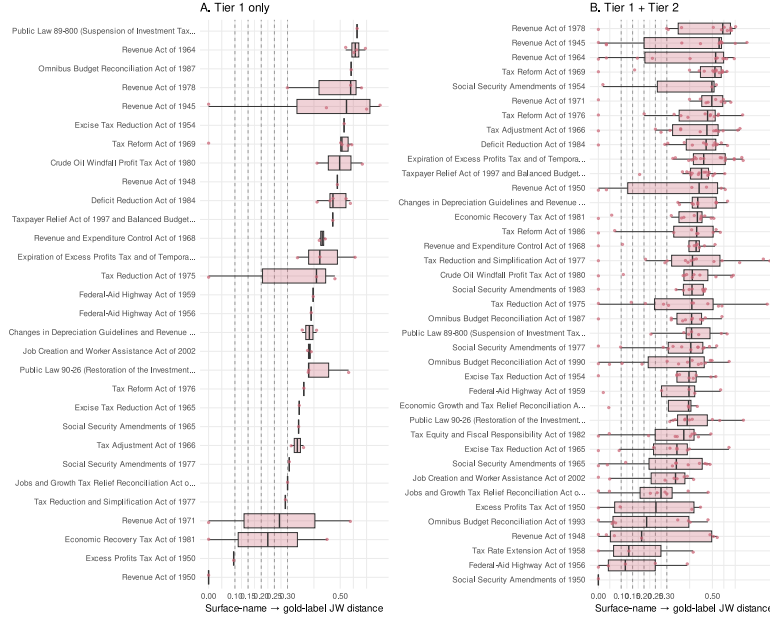


Figure 6: Surface-name $\rightarrow$ gold-label JW distance. Same panels, threshold lines, and visual style as Figure 5; each point is one (measure_name, gold_act_name) pair.

The within-act figures pin down what JW can mechanically do; the ladders below quantify the consequences on the RR-aligned eval the §Framing committed to. Each row is one variant (threshold $\times$ year-blocking); each cell carries a 95% bootstrap CI from resampling the 49 RR acts. Column definitions are the headline four metrics defined in §“Evaluation against RR’s act list”:

- **Coverage** — share of 49 RR acts hit by ≥ 1 cluster under the joint name + year ± 2 gate.
- **Spurious** — share of clusters that match no RR act under the joint gate. The precision-flavoured column the asymmetric-cost framing weights most.
- **Fragmentation** — mean number of clusters claiming each matched RR act (≥ 1).
- **Year alignment** — diagnostic; share of name-matched pairs that also pass year ± 2 .

Two tables — one per name gate — so the keyword-vs-JW gap is visible side by side. Coverage will typically be higher under the JW gate (it accepts more surface forms); the keyword-vs-JW

gap diagnoses how much of the JW coverage is recovery-by-near-canonical-match rather than recovery-by-aggregation.

Table 7: JW single-linkage, keyword gate. Cells: point estimate [95% bootstrap CI]. Rows: 5 thresholds $\times$ {Unblocked, Year ± 2 }. The keyword gate is name-rewriting-immune: a method gets credit only if a cluster member contains an RR-act subcomponent (from `generate_subcomponents()`).

JW thresh-	Cov-	Spu-	Frag-	Year align-	n	Clus-	RR
old	erage	rious	men-	ment	clusters	matched	act/matched
			ta-		ters		
			tation				
Unblocked							
0.10	0.673 [0.531, 0.796]	0.5157 [0.265, 0.979]	0.2512 [1.867, 2.550]	0.6709 [0.635, 0.771]	1638	71	33
0.15	0.633 [0.490, 0.755]	0.4663 [0.168, 0.983]	0.1871 [1.556, 2.182]	0.5659 [0.574, 0.736]	1502	56	31
0.20	0.612 [0.469, 0.735]	0.4970 [0.175, 0.985]	0.1300 [1.300, 1.667]	0.3043 [0.561, 0.722]	1316	39	30
0.25	0.408 [0.265, 0.551]	0.2577 [0.173, 0.990]	0.1350 [1.054, 1.538]	0.1435 [0.325, 0.526]	814	19	20
0.30	0.102 [0.029, 0.184]	0.0994 [0.190, 0.994]	0.0000 [1.000, 1.000]	0.0014 [0.036, 0.179]	180	1	5
Year ± 2							
0.10	0.837 [0.719, 0.939]	0.7148 [0.262, 0.974]	0.2658 [1.958, 2.583]	0.929 [0.845, 0.929]	1707	89	41
0.15	0.837 [0.719, 0.939]	0.7154 [0.168, 0.977]	0.1878 [1.608, 2.125]	0.906 [0.810, 0.906]	1602	74	41
0.20	0.776 [0.652, 0.878]	0.6562 [0.176, 0.980]	0.1605 [1.400, 1.800]	0.868 [0.739, 0.868]	1465	56	38
0.25	0.755 [0.632, 0.857]	0.6362 [0.169, 0.980]	0.1432 [1.257, 1.591]	0.842 [0.692, 0.842]	1170	44	37
0.30	0.306 [0.189, 0.429]	0.1878 [0.172, 0.991]	0.1726 [1.004, 1.444]	0.432 [0.250, 0.432]	544	12	15

Table 8: JW single-linkage, JW-min ≤ 0.30 gate. Same variants and column definitions as Table 7. Continuous distance; accepts near-canonical variants the keyword gate misses. The Coverage gap from Table 7 is the relaxation bonus.

JW thresh- old	Cov- erage	Spu- rious	Frag- men- ta- tion	Year align- ment	n clusters matched	Clus- ters matched	RR acts
Unblocked							
0.10	0.939 [0.877, 1.000]	0.926 [0.895, 0.958]	7.435 [5.651, 7.435]	0.204 [0.161, 0.204]	1638	208	46
0.15	0.918 [0.838, 0.980]	0.933 [0.900, 0.966]	6.385 [4.806, 6.385]	0.194 [0.154, 0.194]	1502	172	45
0.20	0.898 [0.799, 0.980]	0.939 [0.907, 0.973]	5.429 [4.031, 5.429]	0.190 [0.148, 0.190]	1316	140	44
0.25	0.694 [0.572, 0.816]	0.961 [0.935, 0.987]	3.000 [2.095, 3.000]	0.207 [0.147, 0.207]	814	62	34
0.30	0.163 [0.061, 0.265]	0.994 [0.980, 1.000]	1.000 [1.000, 1.000]	0.205 [0.062, 0.205]	180	3	8
Year ± 2							
0.10	0.939 [0.876, 1.000]	0.917 [0.854, 0.981]	8.429 [6.581, 8.429]	0.201 [0.161, 0.201]	1707	233	46
0.15	0.939 [0.877, 1.000]	0.923 [0.869, 0.976]	7.500 [5.871, 7.500]	0.193 [0.155, 0.193]	1602	204	46
0.20	0.939 [0.878, 1.000]	0.926 [0.892, 0.960]	6.828 [5.067, 6.828]	0.189 [0.152, 0.189]	1465	179	46
0.25	0.939 [0.878, 1.000]	0.932 [0.903, 0.961]	5.067 [3.835, 5.067]	0.176 [0.143, 0.176]	1170	132	46
0.30	0.633 [0.519, 0.755]	0.971 [0.951, 0.991]	1.684 [1.347, 1.684]	0.154 [0.104, 0.154]	544	28	31

Reading the JW results

Three patterns from the RR-aligned ladder, focused on the asymmetric-cost shortlist (low spurious, then high coverage):

First, JW’s spurious rate is structurally high. At the precision-friendly operating point ($t = 0.10$, year ± 2), the keyword gate hits 41 of 49 RR acts but only ~70 of 1707 clusters carry a match — the

remaining ~1640 are near-singletons that aggregate nothing. JW under-aggregates rather than over-aggregates, exactly as Figure 5 predicted geometrically. Tightening the threshold lowers cluster counts (and so the spurious rate) but loses coverage. Loosening to $t = 0.30$ collapses everything: 180 clusters total, 1 matching cluster, 5 RR acts covered. There is no operating point on the JW front that combines low spurious with high coverage.

Second, the keyword-vs-JW gap is informative but bounded. The JW gate adds ~5–15 points of coverage on top of the keyword gate at any given threshold, because JW catches lexical near-canonicals (typographic variants, abbreviations of short distance) that don't contain the full subcomponent string. JW single-linkage as a method does not emit RR-style canonical names — it just keeps whatever surface forms it sees — so the relaxation bonus here is mostly recovery of near-duplicates, not recovery-by-renaming. The gap will be much larger under stateful methods (Method 5) that produce a canonical per cluster.

Third, year-blocking continues to help on every variant. Spurious rates drop a few points and fragmentation usually improves; coverage holds steady or slightly improves. The temporal prior remains structural across the eval pivot.

Bottom line for Method 1

JW single-linkage cannot canonicalise this measure pool on its own. The geometric ceiling from Figure 5 is binding: most of the surface-form pairs that ought to merge sit beyond JW distance 0.30, geometrically out of reach for JW at any threshold considered. The RR-aligned ladder confirms it: at every variant the spurious rate stays above 50%, fragmentation is typically 1.5–3, and even the best variant misses ~8 of 49 RR acts under the keyword gate. The gap between *ERTA* and *Economic Recovery Tax Act of 1981* is semantic, not lexical, and JW has no machinery to bridge it.

What JW remains useful for is the role its mechanics actually support: a precision-friendly **blocker** — a cheap pre-filter that catches near-duplicate surface forms (spacing, punctuation, abbreviation expansion) at near-zero false-positive cost. Method 4's hybrid pipeline below uses exactly this property: JW (or a cheap analog) prunes the pair space, and a semantic verifier handles the merges that surface similarity cannot. The load-bearing canonicalisation work, though, has to come from a method that understands meaning — Methods 2–5 are the candidate replacements.

Embedding-geometry probe (F16 vs FP32)

Method 1 closed on the observation that the gap to canonicalise this pool is semantic, not lexical. The next three sections test whether multilingual embeddings can bridge it. Before reading the eval, one preliminary: the Methods 2/3 embeddings come from a community F16 port of `intfloat/multilingual-e5-large-instruct` (`jeffh/intfloat-multilingual-e5-large-instruct:f16`) running locally under Ollama. F16 compresses the reference FP32 weights and could in principle distort the embedding geometry that HDBSCAN relies on. Two cheap probes on the gold eval pool check that the port preserves the semantic structure HDBSCAN needs.

The probes run on a 2×2 grid: **precision** (F16 local vs FP32 reference via DeepInfra) × **eval slice** (full unambiguous pool vs Tier 1 only). The FP32 reference disentangles two distinct failure modes the F16-only probe couldn't separate: "F16 quantization broke the geometry" (FP32 passes where F16 fails) vs "the model itself can't do this task" (both precisions fail). The Tier 1 slice in turn strips Tier 2's permissive IKA tagging, which §"Eval-set noise probe" below shows is genuinely noisy. Reading the four cells together discriminates quantization noise, model-capacity ceiling, and label noise.

Table 9: Embedding-geometry probe on a 2×2 (precision × eval slice) grid. Top-1 NN same-act rate is the share of eval names whose nearest-neighbour by cosine shares a gold act (restricted to acts with ≥ 2 members so the metric is well-defined). Separation = median within-act cosine – median between-act cosine; positive means the embedding preserves the act-level structure. F16 rows come from the local Ollama port (`jeffh/intfloat-multilingual-e5-large-instruct:f16`); FP32 rows from DeepInfra-hosted `intfloat/multilingual-e5-large-instruct`. Same instruction prefix on both.

Pre- ci- sion	Eval slice	Top-1 NN same- act rate	Me- dian within-act co- sine	Me- dian between-act co- sine	Sep- ara- tion	Eval names	Eval acts
F16 (local Ol- lama)	Full pool (Tier 1 + 2)	0.261	0.903	0.898	0.005	206	38
F16 (local Ol- lama)	Tier 1 only	0.308	0.919	0.902	0.017	39	22
FP32 (Deep- Infra)	Full pool (Tier 1 + 2)	0.251	0.903	0.898	0.005	206	38
FP32 (Deep- Infra)	Tier 1 only	0.269	0.919	0.902	0.017	39	22

F16, Tier 1: separation 0.017 (floor 0.05), top-1 NN rate 0.308 (floor 0.40) – **FAIL**

FP32, Tier 1: separation 0.017 (floor 0.05), top-1 NN rate 0.269 (floor 0.40) – **FAIL**

Reading Tier 1 vs Tier 1 keeps the gold-partition noise floor constant on both rows so the F16↔FP32 contrast is apples-to-apples.

Verdict – embedding-model ceiling, not quantization. Both F16 and FP32 fail the same floors, so the issue is not F16 specifi-

cally — `multilingual-e5-large-instruct` itself does not produce the relative cosine spread the floors require on this pool. Two interpretations to weigh: (i) E5-instruct embeddings live in a tight band of the unit sphere by design, and the 0.05/0.40 floors may be inappropriate for this family; (ii) the model is genuinely under-resolving fiscal-act semantics and a stronger or differently-tuned embedding family is needed. Either way, the FP32 migration is *not* the fix. Read Methods 2/3 below as the empirical end-to-end test — if HDBSCAN’s Tier 1 numbers are competitive with JW, the floors were the over-strict reading.

E5-family embeddings always live in a narrow band of the unit sphere, so absolute cosines aren’t crisp — the probe targets the *relative* gap between within-act and between-act pairs, which is the property HDBSCAN cares about. Methods 2 and 3 below continue to read `c0_us_embeddings` (the F16 matrix); the FP32 numbers above are diagnostic only.

Eval-set noise probe: surface-year vs doc-year mismatch

An alternative explanation for a soft F16 verdict — and a noise source the within-act geometry from §Method 1 already hinted at — is that the gold pairs themselves are temporally noisy. If C1 sometimes catches contemporary discussions of acts passed decades earlier (e.g. a 2010 ERP paragraph referring back to the Revenue Act of 1964), the resulting (`measure_name`, `gold_act_name`) pair will look lexically distant — not because the model misclassified, but because the surface form drifts as the act becomes historical context rather than current policy. Figure 7 probes this directly: for each eval tuple whose surface form contains a 4-digit year, plot the gap between the document’s publication year and the year referenced in the surface form against the JW distance to the gold label.

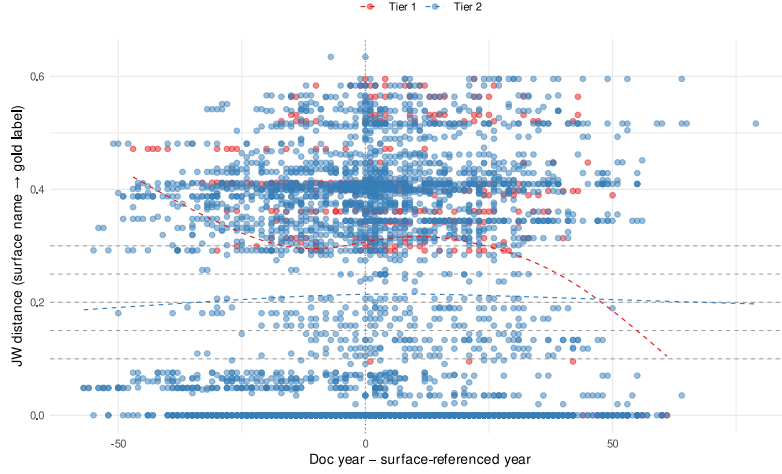


Figure 7: Temporal mismatch in eval pairs. Each point is one (measure_name, gold_act_name, chunk_id) tuple whose surface form contains a 4-digit year. x = doc publication year – year referenced in the surface form (large positive $\Rightarrow$ retrospective reference); y = JW distance from surface form to gold label. Tuples whose surface form has no extractable year are omitted.

Figure 8 repeats the panel on the semantic axis: y is cosine distance between the surface-name and gold-label embeddings (E5-multilingual F16, L2-normalised – so cosine distance = $1 - \text{dot product}$, on the same scale as the F16 probe). The interpretation flips relative to JW: if retrospective references drift *only* lexically (paraphrase) but stay semantically anchored, the cloud should stay flat as x grows; if they drift semantically as well (different terminology or focus when describing the act decades later), the cloud should rise. Reading the two figures together separates the two failure modes.

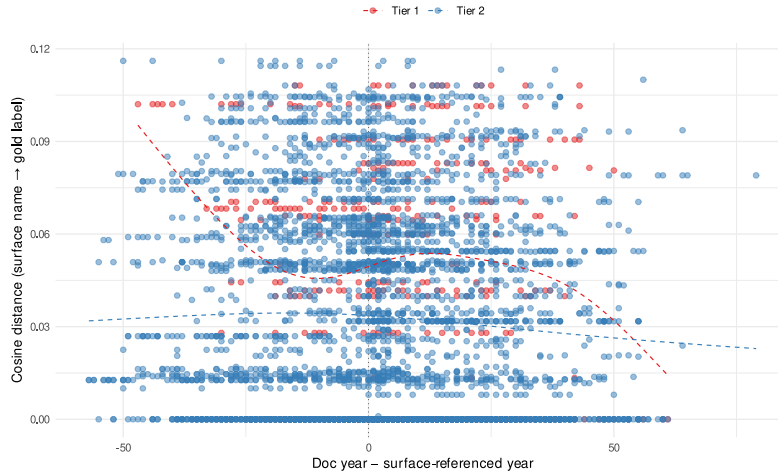


Figure 8: Same panel as Figure 7 but with cosine distance (E5-multilingual F16 embeddings, L2-normalised) on y. Pairs are dropped when either the surface form or the gold label is absent from its respective embedding matrix.

Finally, Figure 9 puts the two distance measures head-to-head on the full eval pool (no year filter): one point per distinct `(measure_name, gold_act_name)` pair, x is the JW distance, y is the cosine distance. The correlation answers whether the embedding is buying us anything JW couldn't already give us — a strong positive correlation would mean E5 is essentially re-deriving lexical similarity, while a weak correlation means the two distance measures disagree on which pairs are similar, and there is genuine semantic signal in the embedding above and beyond surface overlap. Both Pearson (linear) and Spearman (rank) correlations are reported because the JW–cosine relationship is plausibly nonlinear.

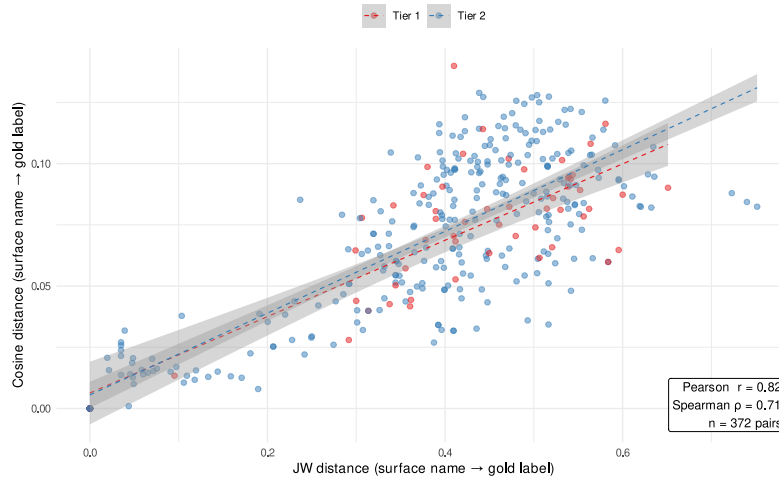


Figure 9: JW distance vs cosine distance for every distinct (measure_name, gold_act_name) pair in the eval pool. Colour marks the cleanest tier in which the pair appears (Tier 1 if at least one chunk pairs the surface form with the gold act via verbatim match). Pairs whose surface form or gold label is missing from its embedding matrix are dropped. Annotated correlations include the full pool; the dashed line is an unweighted linear fit.

Method 2 — Multilingual embedding + HDBSCAN

Switching from string distance to embedding distance is what the within-act geometry from §Method 1 demanded. Method 2 embeds each measure name with `intfloat/multilingual-e5-large-instruct`, L2-normalises the vectors so cosine ranking equals euclidean ranking, then runs HDBSCAN over the resulting distance matrix. Three `min_cluster_size` values are swept — 2 (most permissive), 3, and 5 (close to the median gold-act size). Singleton “noise” points HDBSCAN refuses to merge are kept as their-own-clusters; under the RR-aligned eval they contribute to the spurious rate denominator (singleton clusters with no RR-act match raise spurious; singletons that *do* contain a canonical name still match).

Table 10: HDBSCAN unblocked, keyword gate. Cells: point estimate [95% bootstrap CI]. Rows: 3 `min_cluster_size` settings. Compare to Table 7 — if embeddings buy us what JW couldn’t, coverage should hold and spurious should drop.

<code>min_cluster_size</code>	Cov- erage	Spu- rious	Frag- men- ta- tion	Year align- ment	n clusters matched	Clus- ters mismatched	RR
2	0.592 [0.492, 0.714]	0.492 [0.358, 0.911]	0.583 [0.858, 1.176]	0.0062 [0.654, 169]	30	29	
3	0.408 [0.265, 0.531]	0.645 [0.867, 0.934]	0.675 [1.205, 1.700]	0.0047 [0.429, 181]	28	20	
5	0.449 [0.306, 0.592]	0.825 [0.856, 0.927]	0.832 [1.385, 1.923]	0.0027 [0.500, 206]	36	22	

Table 11: HDBSCAN unblocked, JW-min ≤ 0.30 gate. Same variants as Table 10. The keyword-vs-JW gap diagnoses recovery-by-aggregation vs. recovery-by-canonical-form.

<code>min_cluster_size</code>	Cov- erage	Spu- rious	Frag- men- ta- tion	Year align- ment	n clusters matched	Clus- ters mismatched	RR
2	0.918 [0.835, 0.980]	0.356 [0.222, 0.793]	0.222 [1.067, 2.267]	0.0679 [0.166, 169]	53	45	
3	0.776 [0.657, 0.898]	0.73 [0.796, 0.862]	0.816 [1.591, 2.036]	0.0214 [0.180, 181]	41	38	
5	0.816 [0.713, 0.918]	0.738 [0.267, 0.845]	0.775 [1.082, 2.381]	0.0226 [0.191, 206]	54	40	

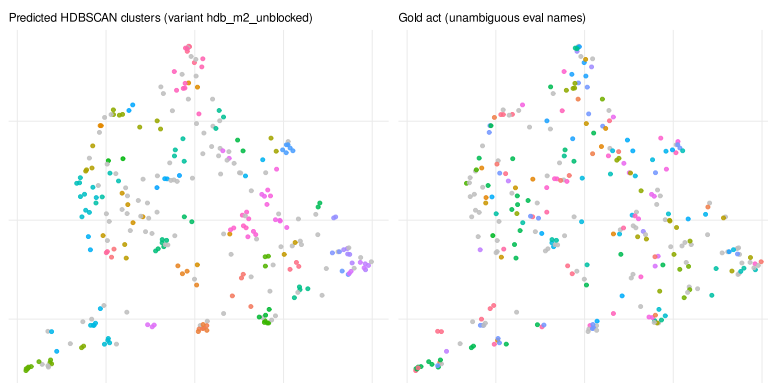


Figure 10: UMAP projection of E5 embeddings. Left: coloured by predicted HDBSCAN cluster (best unblocked variant by Tier 1 F1); singleton/noise names in grey. Right: coloured by gold act for names in the unambiguous eval pool; names outside the eval pool in grey. Same 2D coordinates in both panels.

Reading the HDBSCAN results

The headline comparison is Table 10 against Table 7: if embeddings buy us what JW couldn't, HDBSCAN should hold or improve coverage while collapsing the spurious-rate gap. The UMAP scatter Figure 10 is diagnostic — predicted clusters that overlap a single gold colour on the right panel are correct merges; predicted clusters spanning multiple gold colours are over-merges (note: the gold panel uses `c0_eval_gold_pairs` so the colour palette inherits the gold-pool noise documented in §“Why we pivoted away from pairwise P/R” — read it as a relative-structure visualisation, not as ground truth on which clusters are correct). The §Synthesis at the bottom of the notebook applies the precision-friendly shortlist rule (lowest spurious, then highest coverage) to pick a candidate variant across methods.

Method 3 — Year-blocked variants

Method 3 reruns the same HDBSCAN sweep with the year-window mask from §Method 1: pairs whose chunk years differ by more than 2 are forced to maximum distance and so cannot merge. The prior is that the same act tends to surface in a narrow temporal window, so the mask should reduce cross-decade over-merges without sacrificing coverage.

Partial redundancy with the match gate. The RR-aligned eval already applies a year ± 2 window between cluster representative year and `year_signed`, so a cluster that *would* have over-merged

across decades will fail the year gate anyway and not match the wrong act. Year-blocking at clustering time and year ± 2 at match time are not the same operation, though: clustering-time blocking prevents the merge from happening (the cluster doesn't form), while match-time blocking lets the merge happen but disqualifies the resulting cluster from claiming an act of the wrong year. So year-blocking still helps by reducing the *count* of cross-decade clusters that would otherwise raise the spurious rate denominator. The expected pattern: spurious rate drops modestly under year-blocking; coverage and fragmentation hold.

Table 12: HDBSCAN year-blocked (± 2 at clustering), keyword gate. Same column definitions as Table 10.

min_cluster_size	Cov- erage	Spu- rious	Frag- men- ta- tion	Year align- ment	n clusters	Clus- ters matched	RR not matched
2	0.755 [0.633, 0.857]	0.833 [0.829, 0.886]	1.008 [1.049, 1.182]	0.976 [0.955, 1.000]	38	37	
3	0.673 [0.539, 0.796]	0.807 [0.840, 0.901]	1.403 [1.087, 1.429]	0.878 [0.815, 0.941]	35	33	
5	0.224 [0.122, 0.347]	0.898 [0.890, 0.959]	1.000 [1.000, 1.000]	0.409 [0.178, 0.409]	5	11	

Table 13: HDBSCAN year-blocked (± 2 at clustering), JW-min ≤ 0.30 gate.

min_cluster_size	Cov- erage	Spu- rious	Frag- men- ta- tion	Year align- ment	n clusters	Clus- ters matched	RR not matched
2	0.918 [0.836, 0.980]	0.712 [0.712, 0.783]	1.008 [1.008, 2.519]	0.164 [0.164, 0.212]	56	45	
3	0.898 [0.772, 0.980]	0.754 [0.754, 0.829]	1.592 [1.092, 2.214]	0.173 [0.152, 0.196]	50	44	
5	0.347 [0.225, 0.490]	0.857 [0.857, 0.939]	1.000 [1.000, 1.000]	0.148 [0.083, 0.148]	7	17	

Compare each row to its unblocked counterpart in Table 10 / Table 11. If year-blocking lowers spurious rate at the same min_cluster_size with comparable coverage, the temporal prior

remains structural across the eval pivot and is worth carrying into Methods 4–5 when they land.

Methods 2 + 3 with UMAP dimensionality reduction

The F16 and FP32 embedding-geometry probes above both failed the `separation  $\geq 0.05$` and `top-1 NN same-act rate  $\geq 0.40$` floors on the gold pool. The verdict text considered two interpretations — F16 quantization noise, and the embedding-model ceiling — but did not weigh a third: **high-dimensional distance concentration**. `intfloat/multilingual-e5-large-instruct` returns 1024-dimensional vectors, and the unblocked HDBSCAN above operates on the full-dimensional cosine distance matrix. Density-based clustering is known to degrade above $k \approx 50$ dim through hubness and farthest-to-nearest ratio compression; the standard fix in the BERTopic / Top2Vec neighbourhood is to interpose UMAP between the embedding and the clusterer, collapsing the ambient dim to $\sim 5\text{--}10$ and (optionally) packing within-cluster points tightly via `min_dist = 0`. This section ports that recipe into Methods 2 + 3 and reads the with-vs-without delta against the no-UMAP baseline above.

The sweep grid is the cartesian product of three UMAP hyperparameters with the existing HDBSCAN grid:

- `n_neighbors  $\in \{5, 15, 30\}$` — uwot default 15; the small end emphasises local structure, the large end emphasises global structure.
- `n_components  $\in \{2, 5, 10\}$` — uwot default 2; BERTopic's standard is 5; 10 is the upper end of clustering practice.
- `min_dist  $\in \{0.0, 0.01, 0.1\}$` — uwot default 0.01; 0.0 is BERTopic's tight-packing default; 0.1 matches the existing diagnostic UMAP plot at Figure 10.
- Inner sweep matches the no-UMAP variants: `min_cluster_size  $\in \{2, 3, 5\}$` and `year_window  $\in \{\text{unblocked}, \pm 2\}$` .

That gives 27 UMAP cells $\times$ 6 HDBSCAN cells = 162 variants. Metric "`cosine`" is fixed (the input embeddings are L2-normalised); UMAP is reduced once per UMAP cell and reused across the inner HDBSCAN sub-cells. The seed is fixed at the target level (`run_hdbscan_umap_clusters_grid(seed = 20260526L)`) so the entire sweep is deterministic under `tar_make()`. Distance metric inside HDBSCAN switches from cosine on the raw embeddings to Euclidean on the UMAP output (UMAP does not preserve L2-

normalisation, so cosine on the reduced space is not meaningful); the year-window mask uses a `max(d) + 1` sentinel instead of the cosine 1.0 cap.

Table 14: Best UMAP cell per (min_cluster_size $\times$ year-blocking) sub-cell, keyword gate. Within each sub-cell we sort by spurious_rate ascending then coverage descending and report the winner. Cells: point estimate [95% bootstrap CI]. Compare each row to the corresponding no-UMAP row in Table 10 / Table 12 — if UMAP wins, the cells should sit at lower spurious with comparable-or-higher coverage.

	Block-clustering	min_cluster_size	year-blocking	neighbors	components	dist	Cov-er-age	Spurious	Fragmen-tation	Year-align-ment	n clusters
unblocked	2	5	2	0.010	0.673	0.644	1.182	0.830	104		
					[0.531, 0.796]		[0.721, 0.817]		[1.091, 1.273]		[0.758, 0.906]
unblocked	3	5	10	0.010	0.571	0.597	1.071	0.714	67		
					[0.429, 0.714]		[0.672, 0.791]		[1.000, 1.133]		[0.609, 0.815]
unblocked	5	5	2	0.010	0.429	0.667	1.048	0.512	45		
					[0.306, 0.571]		[0.689, 0.822]		[1.000, 1.100]		[0.409, 0.615]
year ± 2	2	15	5	0.000	0.714	0.658	1.200	0.913	117		
					[0.592, 0.837]		[0.735, 0.829]		[1.091, 1.300]		[0.852, 1.000]
year ± 2	3	5	10	0.000	0.673	0.663	1.091	0.837	92		
					[0.531, 0.796]		[0.728, 0.826]		[1.000, 1.158]		[0.760, 0.926]
year ± 2	5	5	5	0.000	0.612	0.634	1.333	0.800	93		
					[0.469, 0.735]		[0.688, 0.817]		[1.200, 1.471]		[0.714, 0.879]

Table 15: Same shortlist rule on the JW-min ≤ 0.30 gate. The keyword-vs-JW gap is the recovery-by-near-canonical bonus.

Block-clustering	clusters	neighbors	components	dist	Cov- er- age	Spu- ri- ous	Frag- men- tation	Year align- ment	nclus- ters
unblocked	2	5	5	0.100	0.918	0.522	2.111	0.182	113
					[0.837, 0.980]	[0.575, 0.690]	[1.840, 2.379]	[0.158, 0.208]	
unblocked	3	5	10	0.010	0.735	0.433	2.028	0.177	67
					[0.612, 0.857]	[0.463, 0.627]	[1.727, 2.304]	[0.143, 0.209]	
unblocked	5	5	2	0.010	0.612	0.467	1.700	0.170	45
					[0.490, 0.735]	[0.489, 0.667]	[1.412, 1.950]	[0.133, 0.209]	
year ± 2	2	15	2	0.010	0.898	0.538	2.182	0.188	119
					[0.816, 0.980]	[0.588, 0.697]	[1.900, 2.440]	[0.166, 0.214]	
year ± 2	3	5	10	0.000	0.857	0.554	2.048	0.180	92
					[0.755, 0.939]	[0.576, 0.696]	[1.692, 2.385]	[0.151, 0.211]	
year ± 2	5	5	5	0.000	0.816	0.538	2.100	0.174	93
					[0.714, 0.918]	[0.591, 0.710]	[1.740, 2.458]	[0.145, 0.203]	

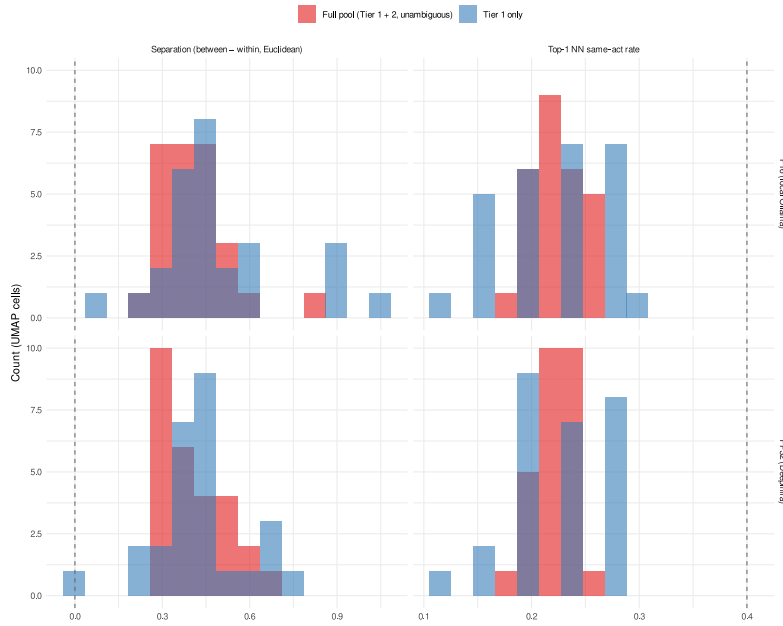


Figure 11: Embedding-geometry probe in UMAP-reduced space — F16 vs FP32, the same model precisions contrasted in Table 9 at no-UMAP. Columns facet the two probe metrics; rows facet embedding precision (F16 = local Ollama community port, FP32 = DeepInfra-hosted reference). The F16 panels (top row) reproduce the earlier UMAP read; the FP32 panels (bottom row) add the missing precision contrast in UMAP space. Each bar counts UMAP cells (out of 27) whose probe value falls in the bin; fill marks eval slice (Full pool = Tier 1 + 2 unambiguous, Tier 1 = verbatim-passage matches only). Dashed lines are the F16/FP32 floors carried over from Table 9: 0.40 for top-1 NN; 0 for separation (Euclidean — the 0.05 cosine floor at Table 9 does not translate). Identical UMAP grid, seed (20260526), and probe logic across the four panels; the only thing that varies between F16 and FP32 rows is the input embedding matrix.

Reading the UMAP results

The headline reading is empirical: across the 162 UMAP variants in `c0_hdbscan_umap_rr_metrics`, **no UMAP variant strictly dominates the no-UMAP HDBSCAN baseline on the coverage/spurious front under either gate**. The shortlist tables above pick the best UMAP cell per (`min_cluster_size` × year-blocking) sub-cell by the asymmetric-cost rule (lowest spurious, then highest coverage); the picks land at comparable spurious rates and comparable coverage to the corresponding no-UMAP rows in Table 10 and Table 12. UMAP-PF (the precision-friendly UMAP winner overall) does cut the singleton share in the cluster-count distribution from

38% (no-UMAP baseline `hdb_m2_yw2`) to 14%, but most of the singletons it removes are noise points that wouldn't have matched any RR act anyway. Two independent diagnostics explain why the front doesn't move.

First, **UMAP improves the aggregate embedding geometry but not the neighbour identity HDBSCAN and the name gates actually use**. Figure 11 shows within-vs-between separation moves dramatically — F16 raw 0.005 → UMAP median 0.379, a $\approx 70\times$ lift, with positive separation in 27/27 F16 cells and 26/27 FP32 cells. Top-1 NN same-act rate barely moves: F16 raw 0.261 → UMAP cells span [0.18, 0.31], and zero of 108 probe cells (54 F16 + 54 FP32 across both eval slices) clear the 0.40 floor. The F16 ↔ FP32 contrast on top-1 NN is flat, which resolves the verdict left open by the no-UMAP F16/FP32 probe in §“Embedding-geometry probe”: the bottleneck is the embedding model, not F16 quantization. UMAP cleans aggregate geometry without changing the literal nearest neighbour, and the literal NN is what HDBSCAN's density-based clustering and the RR-aligned name gates both depend on.

Second, **the upstream gold-pool ceiling documented in §“Why we pivoted away from pairwise P/R” still binds the RR-aligned eval, just differently**. The new framework removes the chunk-tier act_name aggregation noise that capped pairwise recall against `c0_eval_gold_pairs`, but coverage against R&R's 49-act list is still capped by whether HDBSCAN finds ≥ 1 surface form per act that carries enough of the canonical name to satisfy the gates. Tiny clusters that fragment off the abbreviated forms (ERTA, P.L. 97-34) without merging back to the canonical fail both gates regardless of UMAP arrangement. The earlier UMAP draft's third hypothesis (that `n_components` would be the load-bearing UMAP knob) remains empirically wrong on the RR-aligned eval too: no UMAP knob moves the front meaningfully because the front is not where UMAP's lever points. BERTopic-style UMAP-before-density-clustering is therefore not the load-bearing C0 architecture.

Implications for Method 4. The same diagnosis predicts the hybrid LLM pairwise judge will break through *some* of the gap but not all of it. Genuine surface variants (ERTA ↔ `Economic Recovery Tax Act of 1981` ↔ `P.L. 97-34`) that HDBSCAN scattered by 1024-D cosine-distance concentration will resolve once the judge sees the candidate pairs — top-k embedding retrieval should surface them as candidates even at F16 (top-1 may miss but top-5/10 will catch). The judge cannot manufacture canonical names from scratch, though,

so methods that emit canonicals (Method 5) will look better on the keyword gate than methods that only verify pairs (Method 4); the keyword-vs-JW gap in the §Synthesis ladder will surface that distinction.

Method 4 — Hybrid (embedding nearest-neighbours + LLM pairwise judge)

⚠ Phase B pending

The hybrid pipeline (embedding top-k → Haiku YES/NO verifier → union-find) lands once Phase B is built. Expected to dominate the cost/quality frontier: cheap blocker + targeted LLM verification only on the candidate pairs that survive embedding similarity.

Method 5 — Stateful LLM canonical-list build

⚠ Phase B pending

The stateful walker (Haiku reads each doc's measure names, matches to existing canonicals or proposes new ones) lands once Phase B is built. Will include a multi-seed shuffle probe for order sensitivity.

Method comparison ladder

Table 16: Method comparison — best variant per method under the asymmetric-cost rule (lowest spurious_rate, then highest coverage), keyword gate. Methods 4 + 5 are Phase B-pending.

Method	Vari- ant	Cov- er- age	Spu- ri- ous	Frag- men- ta- tion	Year align- ment	m clus- ters	Clus- ters matched	RR matched
JW (best)	hdb_umap_nmd0.10_yw02	0.837	0.948	2.268	0.886	1707	89	41
	vari- ant)	[0.714, 0.939]	[0.960, 0.974]	[1.958, 2.583]	[0.845, 0.929]			
HDBSCAN (best)	hdb_m2_yw02	0.755	0.783	1.108	0.976	175	38	37
	vari- ant)	[0.633, 0.857]	[0.829, 0.886]	[1.040, 1.182]	[0.955, 1.000]			
hdb_umap_nmd0.10_yw02	hdb_m3_umap01	0.671	0.597	1.001	0.714	67	27	28
	SCAN + UMAP (best vari- ant)	[0.429, 0.714]	[0.672, 0.791]	[1.000, 1.133]	[0.609, 0.815]			

Table 17: Same picker, JW-min ≤ 0.30 gate. Methods that re-emit RR-canonical names (Method 5 will, Methods 1-3 don't) gain coverage here that they don't gain under Table 16.

Method	Vari- ant	Cov- er- age	Spu- ri- ous	Frag- men- ta- tion	Year align- ment	m clus- ters	Clus- ters matched	RR matched
JW (best)	hdb_umap_nmd0.10_yw	0.9239 [0.877, 1.000]	0.864 [0.885, 0.917]	7.543 [6.586, 8.429]	0.179 [0.161, 0.201]	1707	233	46
HDBSCAN (best)	hdb_m2_yw	0.9218 [0.837, 0.980]	0.680 [0.714, 0.783]	2.222 [1.903, 2.519]	0.188 [0.164, 0.212]	175	56	45
hdb_umap_nmd	hdb0.10_md0.0135n3_umd0.481	0.9239 [0.612, 0.857]	0.864 [0.463, 0.627]	7.543 [1.727, 2.304]	0.179 [0.143, 0.209]	67	38	36

Three rows now — JW, no-UMAP HDBSCAN, and UMAP+HDBSCAN — picked by the asymmetric-cost rule within each method. Reading across the rows: if UMAP+HDBSCAN dominates the no-UMAP row on coverage with comparable-or-lower spurious, BERTopic-style dim-reduction is the load-bearing improvement. If JW wins on spurious but trails on coverage (the expected pattern given the within-act JW geometry), JW remains a precision-friendly blocker candidate for Method 4 rather than a deployable method on its own. Methods 4 + 5 will extend this table once they land.

Out-of-distribution: Malaysia EN/BM

Phase B pending

Once Phase B is built, the best variant from each method is applied to the Malaysia EN+BM measure pool (`malay_er_cl_measures`); we report the fraction of `malay_er_curated_matches` EN↔BM pairs that land in the same predicted cluster. Methods that rely on surface similarity (JW) are expected to fail here because the language families are different; multilingual embedding methods should hold up.

Cost diagnostics

Table 18: Per-method cost. Methods 1–3 are zero-cost (pure-R or local Ollama). Methods 4–5 will populate from `logs/api_calls.csv` once wired.

Method	Tokens (input)	Tokens (output)	Pilot cost (USD)	Pro- jected SEA cost (USD)	Notes
Method 1: JW single- linkage	0	0	0.00	0.00	Pure-R, no API calls
Method 2: E5- multi- lingual + HDB- SCAN	≈4K	0	0.00	0.00	Local Ollama (jeffh/ intfloat- mul- tilin- gual-e5- large-instruct:f16); embed- dings cached in c0_us_embeddings
Method 3: E5- multi- lingual + HDB- SCAN (year-blocked)	≈4K	0	0.00	0.00	Reuses Method 2 em- bed- dings; year- block- ing ap- plied at the dis- tance- matrix step

⚠ Methods 4 + 5 still pending

Phase B cost diagnostics for Method 4 (Haiku pairwise-judge) and Method 5 (Haiku stateful-builder) will populate from `logs/api_calls.csv` filtered by method tag once those targets run.

Synthesis

i Methods 4 + 5 still pending

A final method + parameter recommendation waits for all five methods to be evaluated. Methods 1–3 (JW, HDBSCAN unblocked, HDBSCAN year-blocked) are now wired against the RR-aligned eval; Methods 4 (hybrid LLM judge) and 5 (stateful canonical builder) remain to land.

Until then, the shortlist below applies the precision-friendly rule from §Framing to each finished method in isolation: among variants with the lowest spurious rate, pick the one with the highest coverage. Compare the JW pick against the HDBSCAN pick to see whether the semantic bridge embeddings provide outweighs whatever lexical-only precision JW retains.

Reading the JW results in isolation

Table 19: JW shortlist under the asymmetric-cost rule (sort: spurious asc, coverage desc), keyword gate. The top row is the most precision-friendly JW variant; the Coverage column says whether that precision came at an unacceptable recall hit.

variant_id	Spurious	Coverage	n clusters	Clusters marked	Reacts matched
jw_t0.10_yw2	0.948 [0.960, 0.974]	0.837 [0.714, 0.939]	1707	89	41
jw_t0.15_yw2	0.954 [0.964, 0.977]	0.837 [0.714, 0.939]	1602	74	41
jw_t0.10_unblocked	0.957 [0.965, 0.979]	0.673 [0.531, 0.796]	1638	71	33
jw_t0.20_yw2	0.962 [0.970, 0.980]	0.776 [0.653, 0.878]	1465	56	38
jw_t0.25_yw2	0.962 [0.969, 0.980]	0.755 [0.633, 0.857]	1170	44	37

Table 20: Same JW shortlist on the JW gate. Coverage typically lifts on the relaxed gate because surface variants share lexical content; the keyword-vs-JW coverage gap is the recovery-by-near-canonical bonus.

variant_id	Spuri- ous	Cover- age	n clu- sters	Clusters marked	Edits matched
jw_t0.10_yw2	0.864 [0.885, 0.917]	0.939 [0.877, 1.000]	1707	233	46
jw_t0.15_yw2	0.873 [0.891, 0.923]	0.939 [0.877, 1.000]	1602	204	46
jw_t0.10_unblocked	0.873 [0.893, 0.926]	0.939 [0.877, 1.000]	1638	208	46
jw_t0.20_yw2	0.878 [0.894, 0.926]	0.939 [0.877, 1.000]	1465	179	46
jw_t0.15_unblocked	0.885 [0.901, 0.933]	0.918 [0.837, 0.980]	1502	172	45

This shortlist is sorted by spurious rate ascending – the column the C0 user should weight most under the asymmetric-cost framing. The top row is the most precision-friendly JW variant; the Coverage column says whether the spurious-rate win came at acceptable coverage cost.

Reading the HDBSCAN results in isolation

Table 21: HDBSCAN shortlist (no-UMAP and UMAP-reduced combined) under the asymmetric-cost rule, keyword gate. [Method](#) marks which family each row comes from.

Method	variant_id	Spurious	Cov- erage	n clus- ters	Clus- ters matched	RR acts matched
hdb_umap_HDB-nc10_md0.010	0.597	unblocked	0.571	67	27	28
SCAN		[0.672, 0.791]	[0.429, 0.714]			
+						
UMAP						
hdb_umap_HDB-nc5_md0.010	0.667	unblocked	0.490	60	23	24
SCAN		[0.667, 0.817]	[0.367, 0.633]			
+						
UMAP						
hdb_umap_HDB-nc5_md0.000	0.619	unblocked	0.510	63	24	25
SCAN		[0.683, 0.810]	[0.367, 0.653]			
+						
UMAP						
hdb_umap_HDB-nc5_md0.003	0.645	unblocked	0.612	93	34	30
SCAN		[0.688, 0.817]	[0.469, 0.735]			
+						
UMAP						
hdb_umap_HDB-nc5_md0.000	0.633	unblocked	0.647	58	21	18
SCAN		[0.690, 0.845]	[0.245, 0.510]			
+						
UMAP						
hdb_umap_HDB-nc5_md0.010	0.644	unblocked	0.688	59	21	19
SCAN		[0.695, 0.847]	[0.245, 0.531]			
+						
UMAP						
HDB-hdb_m2_yw20.783			0.755	175	38	37
SCAN		[0.829, 0.886]	[0.633, 0.857]			
+						
HDB-hdb_m3_yw20.807			0.673	181	35	33
SCAN		[0.840, 0.901]	[0.531, 0.796]			
+						

Table 22: Same HDBSCAN shortlist on the JW gate.

Method	variant_id	Spuri- ous	Cov- erage	n clus- ters	Clus- ters matched	RR acts matched
hdb_umap_HDB-nc10_md0.0100.463_unblocked		0.463	0.612	67	38	36
SCAN		[0.463, 0.627]	[0.612, 0.857]			
+						
UMAP						
hdb_umap_HDB-nc5_md0.0000.494_unblocked		0.494	0.633	63	35	37
SCAN		[0.492, 0.635]	[0.633, 0.878]			
+						
UMAP						
hdb_umap_HDB-nc5_md0.0100.467_unblocked		0.467	0.633	60	32	37
SCAN		[0.500, 0.667]	[0.633, 0.878]			
+						
UMAP						
hdb_umap_HDB-nc2_md0.0100.467_unblocked		0.467	0.490	45	24	30
SCAN		[0.489, 0.667]	[0.490, 0.735]			
+						
UMAP						
hdb_umap_HDB-nc2_md0.0000.482_unblocked		0.482	0.551	56	29	34
SCAN		[0.518, 0.679]	[0.551, 0.816]			
+						
UMAP						
hdb_umap_HDB-nc2_md0.0100.484_unblocked		0.484	0.531	62	32	33
SCAN		[0.516, 0.677]	[0.531, 0.796]			
+						
UMAP						
HDB-hdb_m2_yw20.680		0.918	0.918	175	56	45
SCAN		[0.714, 0.783]	[0.837, 0.980]			
+						
HDB-hdb_m2_unblocked		0.918	0.918	169	53	45
SCAN		[0.722, 0.793]	[0.837, 0.980]			
+						
UMAP						

Same sort rule as the JW shortlist. Reading across the three families (JW above, no-UMAP HDBSCAN, UMAP+HDBSCAN): the deployment pick under each gate is the top row. If a UMAP+HDBSCAN variant displaces no-UMAP HDBSCAN at the top, deploy with UMAP for Malaysia. If JW dominates spurious but trails coverage, JW is the natural blocker candidate for Method 4 — the hybrid

pipeline reconciles the spurious-vs-coverage trade by routing JW's precision-friendly candidate pairs through a semantic verifier.

Romer, Christina D., and David H. Romer. n.d. *A Narrative Analysis of Postwar Tax Changes*.